

Probability - Example Sheet 3-4

1.

Find the median, mean and mode of the following data: 1, 8, 2, 5, 6, 5, 5, 5, 11, 11, 16, 9. Find also the quartiles and the interquartile range.

2.

a) Calculate the mean, variance and standard deviation of the following data: 3, 4, 6, 2, 8, 8, 5, 3, 9, 2.

b) Find the expectation (expected value) of a random variable uniformly distributed over the values 1, 2, 3, ..., n . Prove (by induction) that the variance is $\frac{(n+1)(n-1)}{12}$.

3.

We are firing at a target until we hit it (hitting it with probability 0.05 at each trial). What's the average number of bullets (the munition) we need for hitting the target once?

4.

A balloon manufacturer claims that 95% of his balloons will not burst when blown up. If you have 20 of these balloons to blow up for a birthday party:

a) what's the probability that none of them burst when blown up?

b) Find the probability that exactly two balloons burst.

There are 17 children coming to the party but you decide to blow up all 20 balloons as some spares might be helpful.

c) What's the probability that there are enough balloons?

5.

Flaws in a certain brand of tape occur at random and at an average of 3 per 400 meters. Assuming a Poisson distribution for the number of flaws in a 400 meter roll of tape,

a) find the probability that there will be at least one flaw. (Hint: the parameter of Poisson distribution is equal to its expected value.)

b) Show that the probability that there will be at most 2 flaws is 0.423.

c) In a batch of 5 rolls, find the probability that at least 2 rolls will contain fewer than 3 flaws.

d) What's the average number of flows in 100 meter of tape? What's the probability that there are no flows in 100 meters?

6.

In a fictive dictatorship, the "Big Brother" gives the following order:

a) For giving soldiers to the homeland, each family is obliged to give birth to a son.

b) To avoid population explosion, no more child is allowed after the first boy.

How will these two orders of the Big Brother change the normal 1:1 boy-girl ratio (if everything happens according to the rules)?

7.

a) Prove that the expectation of a Binomial random variable (with parameters n and p) is np .

(Distribution of X is Binomial if $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, where $k = 0, 1, 2, \dots, n$.)

b) Prove that the variance is $np(1-p)$.

(You can use the identity $Var(X) = E\{(X - \bar{X})^2\} = E(X^2) - \{E(X)\}^2$.)

8.

a) Prove that the expectation of a Geometric random variable (with parameter p) is $\frac{1}{p}$. (Distribution of X is Geometric if $P(X = k) = (1-p)^{k-1} p$, where $k = 1, 2, 3, \dots$)

(Hint: use rules of differentiation in the proof.)

b) Prove that the variance is $\frac{1-p}{p^2}$.

9.

a) Prove that the expectation of a Poisson random variable (with parameter λ) is λ . (Distribution of X is Poisson if $P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$, where $k = 0, 1, 2, 3, \dots$) (Good to know that $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$.)

b) Prove that the variance is λ .

c) Prove that the Poisson distribution can be reached as a limit of Binomial distributions, where the n parameter of the Binomial distribution tends to infinity and np remains constant (so p must tend to 0). The parameter of the Poisson distribution will be $\lambda = np$. (Hint: you need that $(1 + \frac{k}{n})^n$ tends to e^k .)

10.

Let the cumulative distribution function of X (denote it by F) be continuous and strictly monotonously increasing for those x values where $0 < F(x) < 1$. Prove that the random variable Y defined by $Y = F(X)$ is uniformly distributed over the interval $[0, 1]$.

(Hint: prove, using the definition of the cumulative distribution function as a probability, and transforming the event whose probability we try to calculate, that the cumulative distribution function of Y takes the value y for any $y \in [0, 1]$.)

Reversed direction: find the way to generate values of a random variable X with any given distribution function $F(x)$, if we can easily generate random numbers uniformly distributed over $[0, 1]$.